

AI SOFTWARE DEVELOPMENT INTERNSHIP

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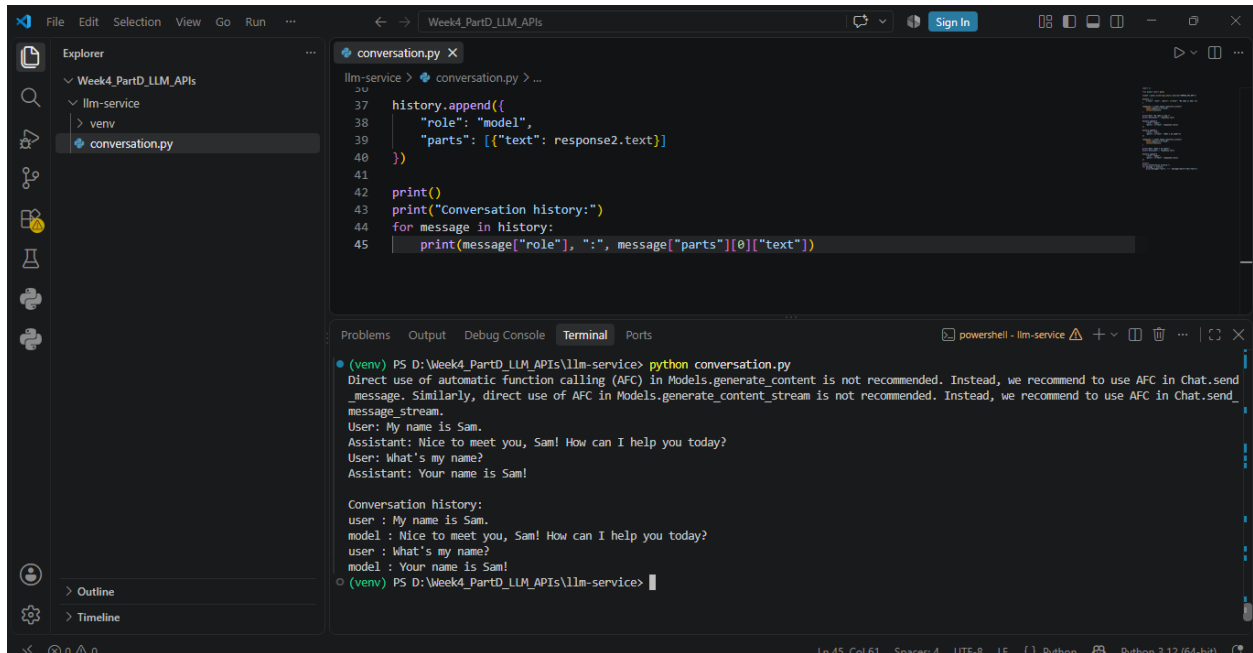
Week 4 — Detailed Task Sheet JWT Authentication (.NET + Angular) • FastAPI
Foundations LLM APIs In Depth • Prompt Engineering • Git — Level Up

PART D — LLM APIs IN DEPTH

Practice Tasks

1. Build a small multi-turn console chat loop that keeps appending to the history list, so the model correctly remembers something you told it two messages ago.

Output:



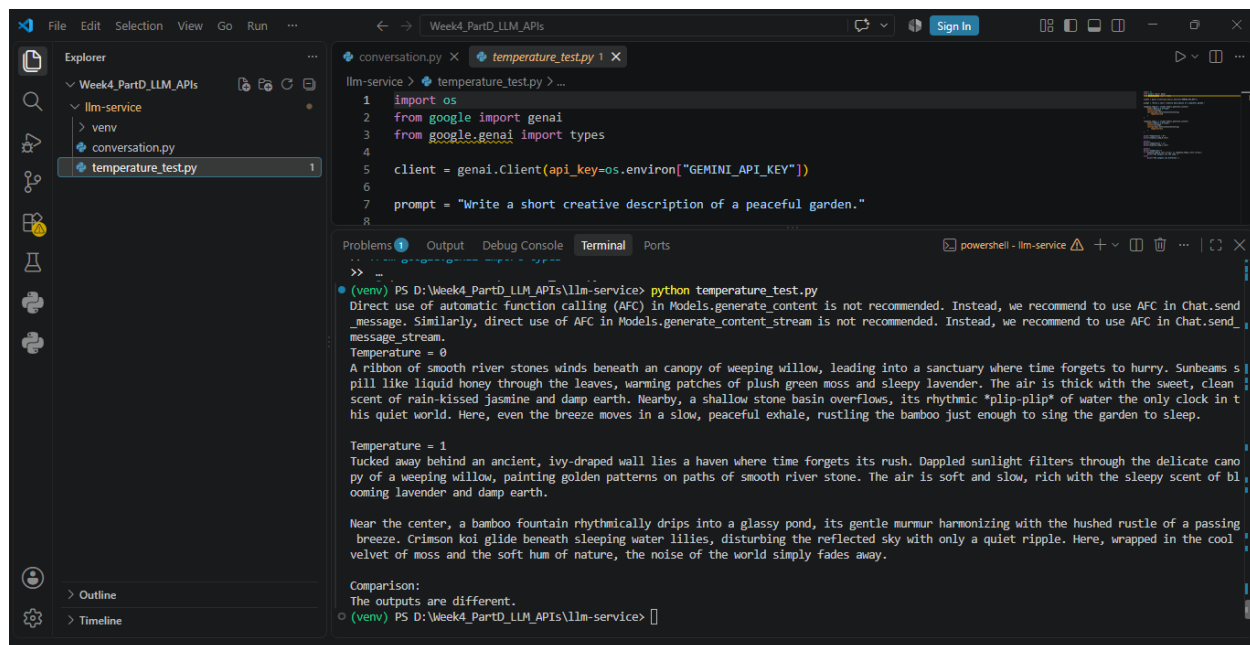
```
llm-service > conversation.py ...
37 history.append({
38     "role": "model",
39     "parts": [{"text": response2.text}]
40 })
41
42 print()
43 print("Conversation history:")
44 for message in history:
45     print(message["role"], ":", message["parts"][0]["text"])
```

```
(venv) PS D:\Week4_PartD_LLM_APIs\llm-service> python conversation.py
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.
User: My name is Sam.
Assistant: Nice to meet you, Sam! How can I help you today?
User: What's my name?
Assistant: Your name is Sam!

Conversation history:
user : My name is Sam.
model : Nice to meet you, Sam! How can I help you today?
user : What's my name?
model : Your name is Sam!
(venv) PS D:\Week4_PartD_LLM_APIs\llm-service>
```

2. Run the same prompt twice with temperature=0 vs temperature=1 and compare how much the output changes.

Output:



3. **Get the model to return valid JSON for the book-summary use case, parse it with `json.loads`, and print just the genre field.**

Output:

